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Gas Turbine Combustion Design, Technology and Research Current Status and Future Direction

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Significant advances have been achieved in the turbopropulsion gas turbine combustion products in regard to operating conditions; combustion system length and volume; performance, operability, gaseous and smoke emissions. Our basic understanding of the critical combustion processes has improved considerably due to the advances made in the measurement technology, numerical simulation and resulting increase in model accuracy. This has in turn benefited the combustion design process which now uses routinely both the approximate and CCD-based methods unlike the old "cut-and-try" design practice. The recently increased collaboration between industry and the combustion research community needs to be further strengthened in order to quickly transition the fundamental know-how to better design tools, improve predictability, and reduce cycle time between "idea-and-product" introduction.

Introduction

Gas turbine combustion design has come a long way in regard to heat release rates, operating pressures (exceeding 40 atm) and temperatures (3000 F in commercial technology engines and near-stoichiometric in high-performance military engine demonstrators), durability, and combustor length. Single annular combustors (SAC) are being pushed to the lowest possible levels of NOx consistent with the other propulsion combustion system requirements.

The dual annular combustors (DAC) have been certified for low emissions turbopropulsion engines that produce an even lower levels of NOx emissions during the landing-takeoff cycle (LTO) as well as for typical flight missions.
The triple annular combustion system for dry ultralow NOx (with potential exhaust NOx levels down to single digits) has been successfully introduced as an alternate to the wet low-emissions control technology for industrial application of aeroderivative gas turbines with nominal pressure ratios of 30 or higher.

Empirical/analytical combustor design approach has become industry's standard for advance technology, new product introduction and current product support activities.
Nonintrusive measurement techniques have made significant progress that has helped create an extensive benchmark quality database for assessing the quality of current turbulent combustion models and providing direction for further improvements.

Extensive validation activities on the k-ε based turbulence models, fast chemistry or finite-chemistry laminar flamelet concepts with specified PDF have established the limitations of these models. This has in turn given the motivation for working on the advanced second-order closure turbulent combustion models; conditional moment closure modeling; Monte-Carlo scalar and/or joint velocity/scalar PDF models; random vortex and large eddy simulations for practical flows.

Description of Modern Combustors

Figure 1 gives a cross-section of the CF6 gas turbine engine combustion system that is used on a number of wide-body commercial aviation airplanes. This engine is produced in several models with the sea-level standard-day static take-off (SLSTO) thrust rating between approximately 175 and 300 kN. The corresponding range of the engine SLSTO pressure ratio (Rc) is 25 -33. Figure 2 presents an extended detail on the combustion system of the GE/SNECMA CFM56 engine with the SLSTO thrust and Rc ranges of approximately 80-150 kN and 21-31, respectively.
As shown, the combustion system consists of several components including a pre-diffuser, casing, fuel nozzles and manifold(s), ignition system, dome, inner and outer liner walls. The turbopropulsion gas combustion systems are designed, developed and certified to meet several design requirements including performance, reliable operation over the entire flight envelope and structural durability in addition to minimizing the undesirable pollutants (NOx, CO, hydrocarbons, HC and smoke). Because of the continuous improvements made over the last 25 years on the single

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annular combustors (SAC), the smoke, CO and HC emissions have been reduced by an order of magnitude in addition to approximately 20% reduction in NOx. Significant reductions in NOx emissions can be achieved by using the dual-annular combustors (DAC), as shown in Fig. 3 for the GE/SNECMA CFM56. Refer to Danis, Burrus and Mongia [1] for more details on Figures 1 through 3.

Many of the industrial gas turbines for power generation, co-generation, and natural gas pumpers use water/steam injection to reduce NOx emissions at high power. Aero-derivative engines using dry-low emissions control technology, DLE have been installed recently at a number of customer installations. For example, the LM6000 DLE engine with a triple-annular combustion system, as shown in Fig. 4 (refer to Joshi et al [2] for details) has been operating in the field since January 1995.

Emissions of Turbopropulsion Combustors

All large commercial propulsion engines are required to measure the exhaust emissions over the landing takeoff (LTO) cycle, as specified in Annex 6, Vol. II of the ICAO Chicago Convention. The measured emissions at idle, approach, climb-out and take-off conditions are generally presented as characteristic values (g/kN) for the LTO cycle. The LTO NOx values for several old and newly-certified propulsion engines are shown as a function of the sea-level static takeoff (SLTO) overall engine pressure ratio (OPR) in Fig. 5. The old NOx standard of 40+2OPR (g/kN calculated over the LTO cycle) has recently been replaced by 32+1.6OPR (effective December 31, 1995). Here OPR denotes the rated sea-level static, standard-day take-off pressure ratio. The slopes of 2.0 (for the old standard) and 1.6 (for the new standard) are allowed in recognition of the higher combustor inlet temperatures and the attendant flame temperatures for the high-pressure ratio energy-efficient turbofan engines. As discussed later, the measured slope values can be significantly different from 1.6. It is a common practice to compare the LTO NOx at a nominal pressure ratio of 30. Therefore, the maximum LTO NOx limits established by the old and new standards at 30 pressure ratio are 100 and 80 g/kN-LTO cycle, respectively. The characteristic NOx for the CF6 (Fig. 1), CFM SAC (Fig. 2) and DAC (Fig. 3) interpolated for 30 pressure ratio are 50.7, 67.0 and 42.5 g/kN -LTO cycle, respectively.

It is a common practice to rank the competing emissions technology options by comparing the NOx levels, and the trends of the NOx/CO and NOx/HC curves. For example, Fig. 6 compares the characteristic values of NOx and CO for all the engines presented in Fig. 5. Other than the old technology engines, all state-of-the-art commercial propulsion engines have CO and HC emissions less than the ICAO standards, namely 118.0 and 19.6 g/kN-LTO cycle, respectively.

Fig. 7 presents the LTO NOx emissions of the CF34 single-annular combustor (SAC), CFM SAC, the CFM dual-annular combustor (DAC) and the GE90 DAC as compared to the 1996 ICAO standard, and the goals of a recently initiated Advanced Subsonic Transport (AST) combustion technology programs for application in the regional and wide-body airplanes. The slopes of the curves for the CFM SAC, CFM DAC and the GE90 DAC are 3.3 ($R^2 = 1.0$), 2.3 ($R^2 = 0.95$) and 2.9 ($R^2 = 0.97$), respectively, as compared to 1.6 allowed in the 1996 ICAO standard. Since the fuel-air mixing technologies of both the CFM and GE90 DACs are comparable, the LTO NOx of both of these combustor can be best-fitted with a single linear curve with a slope of 3.1 and $R^2 = 0.99$.

The low-emissions technology evolution can be best illustrated by comparing NOx (for example, as in Fig. 7), the NOx/HC and NOx/CO characteristics, as shown in Figures 8 and 9. The CF6 family of engines (with the rated take-off pressure ratio and thrust ranges of approximately 25-33 and 175-300 kN) have three different levels of emissions technologies (identified as the CF6-6/-50, CF6-80C and CF6-80LEC) developed during the last 20 years. The most recently certified third-generation low-emission technology (the CF6-80LEC) has significantly lower HC and CO emissions (2.7 and 29.3 g/kN, respectively) in addition to the low characteristic NOx level of 50.7 g/kN for a nominal pressure ratio of 30. Through continuous improvements in the CF6 single annular combustor design, approximately 20% reduction in NOx has been achieved in addition to more than 90% and 50% reduction in unburned hydrocarbons and CO emissions, respectively.

In order to reduce emissions beyond what has been achieved to-date by the single annular combustors, a number of advanced concepts have been developed starting with the NASA sponsored Experimental Clean Combustor, ECC program [3,4]. The CFM56-5B engine model certified recently with a low-NOx dual-annular combustor emits 35% lower NOx over the LTO cycle compared to the single annular combustor in a similar engine model. In addition, the DAC's altitude cruise NOx emission index is approximately 50% lower than that of SAC. The low-emissions technology developed under the ECC and subsequent Energy Efficient Engine, E3, programs was applicable to large turbopropulsion engines of nominal pressure ratio of 30. Its application to the GE90 of nominal pressure ratio of 40 has been quite successful in that the NOx versus pressure ratio, HC/NOx and CO/NOx characteristics are identical for both the CFM56 and the GE90, as shown in Figures 7, 10 and 11, respectively.

The multi-zone combustion systems (e.g., dual-annular, triple-annular DLE [2], and axially-staged [4 and 5]) pose a number of design challenges that are being addressed as part of new product introduction activities. The DAC engines' entry into revenue service has been considered successful by its launch

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customers. The DAC's HC/NOx are slightly better than that of the CFM56-5B SAC, as shown in Figures 10. The CO/NOx characteristics for both the SAC and DAC are comparable as shown in Fig. 11. The axially-staged combustion systems also pose a number of design challenges. Its NOx/CO characteristics are comparable with its counterpart SAC, identified as Phase I in Fig. 12 reproduced from Segalman et al [5]. Based on the design and development experience on the SACs (e.g., Figures 8 and 9 for the CF6 engine series), the next-generation DAC engines will have significantly lower CO and HC in addition to lower NOx demonstrated in the first-generation DAC product. This is further elaborated in the following section for the dry-low emissions (DLE) engines which emit lower NOx, CO and HC emissions compared to the wet-control technology SACs.

Emissions of Aero-derivative Industrial Engines

The water/steam injection (wet) control technology is generally used for meeting the local stringent emissions standards for industrial gas turbine applications. In order to meet 25 PPM NOx emissions (referenced to 15% O2 content in the exhaust) requirements for a high-pressure aero-derivative gas turbine engine (nominal pressure ratio of 30 and typical 225 PPM NOx with NG and no-control), one is required to achieve approximately 90% reduction in NOx within acceptable levels of CO emissions. Many installations allow 42 PPM NOx for Diesel fuel (DF-2) fired gas turbine installation as compared to the uncontrolled NOx (for a nominal 30 pressure ratio engine) of 425 PPM, as shown typically in Fig. 13.
The dry-low emissions (DLE) control technology is preferred over the wet control technology because of several reasons including life-cycle cost and water conservation. The DLE combustion system described by Davis and Washam [6] is being used in several heavy-duty combustion turbines. Many DLE combustion systems [7-11] have since been developed and installed in customer sites including the highest pressure ratio LM6000 DLE combustion system described by Leonard and Stegmaier [9] and Joshi et al [2]. This combustion system (as shown in Fig. 4) uses a multi-stage combustion concept with enhanced lean-blowout, blow-out avoidance control logic and acoustic energy absorption/modulation devices to achieve the ultra-low emissions, performance, and durability objectives.

A typical comparison between the uncontrolled NOx and DLE with NG is shown in Fig. 13. As compared to 225 PPM NOx at the 100% rated power with an uncontrolled diffusion flame combustor, the DLE produces less than 25 PPM and the attendant 90+% NOx reduction. The corresponding NOx-CO emissions characteristics are compared in Fig. 14. The premix DLE combustion system with a five-stage combustion staging logic involving the three partially-separated combustion domes meet all the design requirements including significantly reduced NOx and CO emissions for the power demands above 50% of the rated output.

Approximate Design Calculation Methods

In order to meet the challenging design requirements of new environmentally acceptable energy efficient gas turbines, one needs to develop the following two levels of design tools.
(1) A simplified computationally cheap analysis methods that utilizes the state-of-the-art empirical, semi-empirical and semi-analytical understanding of combustion processes that impact emissions, performance, operability and life characteristics. The validated correlations are used for the preliminary design, trade studies, detail design, data reduction and analysis (DRA) of existing and future design concepts.

(2) Anchored computational combustion dynamics (CCD) modeling approach based on refined empirical turbulence model constants and boundary conditions targeted for the design system applications in current and growth engine combustors. This topic will be covered later in this paper.

A number of semiempirical expressions have been proposed over the years for correlating NOx, CO and unburned hydrocarbon emissions of gas turbines. A "single reactor" modeling approach has been used by Lefebvre [12] to correlate emissions data from the several military engine combustors, but required design specific ("arbitrary") constants. In an alternate approach, Mellor and his associates [13-14] have used the characteristic-times concept to correlate emissions. Rizk and Mongia [15] extended Lefebvre's approach by including the effects of spray evaporation and mixedness on emissions. Rizk and Mongia [16] proposed semianalytical expressions for NOx, CO and HC based on a network of seven reactors and a detailed propane-air combustion mechanism. With this approach they showed that different expressions are needed for the NOx formation rates in the primary zone when φ is below or above 1.08. Moreover, a different expression was proposed for the intermediate zone.
Lefebvre's correlations [12] have been developed without making explicit use of the airflow distribution around the combustors. Rizk and Mongia [16] have utilized the calculated airflow distribution in their formulation; but their approach has been verified for detailed rig emissions database on only one experimental combustor hardware.
In order to develop the first-generation DRA tools for emissions, Danis, Pritchard and Mongia [17] used the calculated airflow distribution and extended Lefebvre's approach [12] for several modern certified commercial

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propulsion engines including the CF6-80C2, LEC SAC, and the CFM56 SAC and DAC shown in Figures 1 through 3. Fig. 15 shows comparison between the predicted and measured NOxEI for the four combustors. Overall the agreement is good. The agreement for the high-power range (defined arbitrarily as NOxEI ≥ 10 g/kg, which generally covers the operating conditions of approach and above) is within ± 10%. for most of the points. The level of agreement between the data and the semi-empirical model needs to be improved for the low- to intermediate-power operation, however.

Figures 16 and 17 show comparison between the predicted and measured COEI and HCEI for the four combustors. If one pays attention to COEI and HCEI values equal to or greater than 1.0, the overall agreement between prediction and data is quite good. However, the CO model under-predicts up to 50% at lower values. The hydrocarbon predictions are within ± 30%.

Another example of DRA is the semi-analytical formulation and validation of an airblast nozzle model by Rizk and Mongia [18]. The authors used semi-analytical expressions for liquid preparation, break-up, secondary-atomization, velocity field and spray transport processes. The semianalytical model approach was calibrated with six different atomizers (that included both the pre-filming and plain jet types) and six liquids including kerosene, JP-5, Gas Oil, MIL-C-7024, water and water/alcohol mixtures. Fig. 18 shows typical comparison between the measured and predicted Sauter Mean Diameter. This approach is considered a significant improvement over the state-of-the-art empirical modeling techniques reviewed by Lefebvre [19].

Qualitative Help from CCD in Design Process

The conventional design approach makes extensive use of database, empirical and semi-empirical design correlations, simplified calculation methods (mostly quasi-1D), correlative multi-dimensional heat transfer and stress analysis, coupled with an extensive use of the single-cup, sector and annular test rigs, and development engines to design and develop the combustors. This is followed by well-defined engine certification process. The limitations of this mostly empirical design process has been well-known. The first published report wherein extensive use of multi-dimensional computational dynamics (CCD) was made to augment the conventional empirical approach is by Mongia et al [20]. This empirical/analytical design methodology has been used extensively during the last 20 years [21-22] in the design and demonstration of the 15 advanced technology combustors, four demonstrators, three certified product engines, and one small turbine augmentor.

In addition, CCD has helped in improving our basic understanding and providing guidance for addressing design, development and product-sustaining issues in several engines including the TFE731, TPE331, GT601, 501-K34, 570, T56-A427, T406/GMA2100, GMA 3007, CFM56-5B DAC1 and DAC2, CF6-80C2, CF6-80LEC and GE90DAC1 and DAC2.

Even though the empirical/analytical design methodology is now used widely in the gas turbine industry [20-27], significant advances need to made in the modeling of the critical combustion processes as explained in the following paragraphs.

Limitations of CCD

A number of articles have compared model predictions with data for the flows relevant to gas turbine combustors [21, 28-48]. From these investigations, one can conclude that the current models are not accurate enough to give the level of accuracy the combustion designers would like to have as given in Table I. Even though a number of model improvements have been made during the last 15 years, the models used in gas turbine combustion design system generally use the standard k-ε model, a two-step kinetic scheme with a variant of the modified eddy break-up model [20] and/or fast chemistry with specified PDF, stochastic spray model etc. There are several critical processes that need to be modeled for gas turbine combustion. We have selected the following three examples to quantify a typical level of disagreement for the models currently used in the design system:

1) Flow field inside and downstream of a "production" airblast nozzle.
2) Modeling of the CFM56 swirl cup shown in Fig. 2
3) Dilution jet mixing

The ability to predict fuel injector flowfield and atomization quality is very critical since the fuel nozzle directly influences a number of aerothermal performance parameters, emissions levels, wall temperature levels and gradients. Nikjooy et al [45] conducted a systematic experimental and analytical study for nozzle-swirler interaction. Their work showed that the Reynolds stress model of the 80's is only slightly better than the k-ε model when the calculations start with the profiles measured downstream of the airblast injector. However for design application, where one has to start calculation from within the airblast passages, the agreement between measurements and predictions is very poor [49]. Consequently, the comprehensive fuel injector modeling activities conducted over the last ten years have not provided any reliable design tool for optimizing the airblast nozzle/dome interaction. Fig. 19 is a typical illustration for the lack of agreement between the measured and predicted radial profiles of SMD from a comprehensive airblast injector model

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[46] at an axial station 50 mm downstream from the nozzle exit.

Experimental investigations on swirl cups (e.g., Fig. 2) as discussed later have shown the complexity of the critical processes. Tolpadi, Burrus and Lawson [47] used a complex fine-grid network of 144 x 93 body-conforming grids along with a second-order accurate QUICK numerical scheme to predict flow-field inside and downstream of the CFM swirl cup. As shown in Fig. 20, the agreement between predicted and measured profiles of the three velocity components 17 mm downstream from the swirl cup exit plane is at only qualitatively good.

The dilution jet mixing process is considered critical for achieving the required burner exit temperature quality including radial profile and pattern factor. Holdeman [50] has given an extensive review of the available experimental database, and simplified correlations in dilution jet nixing, in addition to somewhat limited review on the multi-dimensional computations. Although the k-ε model-based 3-D calculations agree qualitatively with data, one needs to make significant improvement in numerics, turbulence modeling and resolve boundary condition related issues in order to achieve acceptable results.

Perhaps the best illustration for the lack of a good jet mixing model can be made by referring to the experimental and analytical investigation reported by McGuirk and Spencer [48]. They used two concentric tubes of 154 and 94 mm diameters, respectively, with a single row of six dilution orifices located 90 mm downstream from a reference station. They made LV measurements for a constant jet momentum ratio of 25. 3D elliptic flow calculations were made for both the annulus and tube flowfields by using cylindrical polar and curvilinear coordinate systems. With a coarse grid network of 48 x 40 x 16 (axial x radial x circumferential direction, respectively), approximately 75 nodes simulated the half dilution orifice. With a finer mesh of 74 x 110 x 20 nodes, the half dilution orifice was simulated by 200 nodes. The half circular orifice was simulated by stair-step (castellated) approximation and exactly by the cylindrical polar and curvilinear coordinate systems, respectively. The predicted effect of the four simulations (fine and coarse grids, castellated and exact simulation of the dilution orifice) on the jet mixing characteristics were significantly different, as shown in Fig. 21 for the axial velocity profiles along the tube centerline in-line with the dilution orifice. The best and worst agreements between the data and predictions were obtained with the most approximate (coarse castellated) and exact (fine fitted) simulations, respectively.

Mongia [51] has presented several other examples of both the reacting and non-reacting flows relevant to gas turbine combustors that clearly show that the turbulent combustion models and the numerical simulations used in the state-of-the-art combustion system design codes are good enough only for providing qualitative guidance during design. In order to get more reliable quantitative guidance as summarized in Table I, industry needs to adopt and refine the most recent turbulent combustion modeling and simulation techniques in the design process, as discussed later in this paper.

Quantitative Help from CCD in Design Process

Due to the limitations of the state-of-the-art turbulent combustion models as pointed out by many including references [20-22, 39, 45, 49-51], a number of modeling techniques have been proposed [1, 52-61] to achieve the quantitatively accurate predictions of performance and emissions parameters to the level of accuracy required by the combustion designers. All of these approaches have been useful for correlating and analyzing quantitatively existing combustor designs. With the cumulative experience with these approaches, it is now possible to use the state-of-the-art turbulence combustion model through the anchoring process as described in Reference 1, to design a new combustor with more confidence than the empirical/analytical design approaches similar to that of Reference 20.

As described by Danis, Burrus and Mongia [1], the anchored CCD was first calibrated with one of the five modern turbopropulsion combustors. Fig. 22 shows a typical comparison between predicted and measured average radial profiles of temperature at the burner exit plane. The anchored models were then run for the other four combustors. The predicted results correlated well with measured NOx, CO, HC, LBO and exit temperature quality data, demonstrating a broader applicability of the anchored method. The models were also used for designing a new combustion concept. The pre-test predictions agreed well with the sector rig data from a development hardware, as shown in Fig. 23 for NOx (from sea-level idle to the maximum take-off power), and CO and HC at the simulated idle (for the constant inlet pressure, temperature and air flow rate) as a function of fuel/air ratio.

Table II lists the average error, defined as $\sum_n$ Abs (% Error)$_i$ / n , for the whole power range from maximum to idle. In some cases, it might be more meaningful to compare the differences between predictions and data in terms of EI's instead of percentages. From the first entry in Table II for Combustor 5, one can see that even with anchoring the average errors for the whole power range are 25%, 69% and 62% for NO, CO and HC, respectively. Most of the error in NOx is associated with the low-power range, whereas the errors for CO and HC are very high in the high power range. If high power region is arbitrarily defined as the region with NOxEI greater than 10.0 g/kg, then the average error for NO is only 8.0%. The corresponding errors for CO and HC are only 1.5 and 0.4 g/kg, respectively. Similarly, for the low-power region where the NOx contribution to the total emission is not significant,

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the average error for CO is only 4.0%. However, the corresponding error for HC is very high, 65%. The agreement for the lean blowout (LBO), radial profile factor (RPF), and peak-temperature factor (PTF) is good.

Combustor 5 is a high-pressure dual annular combustor with a "near-stoichiometric" primary zone. If one can use the anchored CCD for correlating data from other combustors, not previously used in the anchoring process, and get good agreement with data, this would be a useful tool for use in the combustion design process. To check the feasibility of this approach, the anchored CCD was used to make predictions for three single annular combustors (Combustors 1, 3 and 4), and a lean dual-annular combustor (Combustor 2). These runs were made with the same procedure, and without modifying any of the model constants and/or boundary and inlet conditions established for Combustor 5. The anchored process gave good agreement with the data. from existing combustors, as shown in Table II.

The last entry in Table II is the average errors from the semi-empirical "single-reactor" modeling correlation described by Danis, Pritchard and Mongia [17], refer to Figures 15-17. From here it is seen that the original intent of the anchoring CCD has been achieved in that the multi-dimensional modeling predicts emissions levels with the confidence levels comparable with that of the semi-empirical models.

Advanced Diagnostics for Design

The first extensive data collection from practical combustors for model calibration used intrusive techniques [20-21] for detailed measurements inside a number of rig combustor configurations as well as several "production" combustors. Only a few nonintrusive measurements even at the combustor exit plane [62-63] are available for conducting model calibration. However, a number of advance diagnostic techniques have been used recently for studying several combustion subcomponents and the critical processes. For example, the detailed measurements on model combustors [64-69] have been used extensively for model calibration. Similarly, one can learn a lot from the measurements for the flow over a backward facing step [70], swirling flow in a can research combustor [71], the vortex break-down of a strongly swirling confined flow including the effect of density ratio [72-73], the impact of shroud on the nozzle flow field [74], and finally the effect of the initial profiles on the swirling jet expansion including vortex breakdown [75].

As summarized by Bachalo [76], the advanced diagnostic techniques have been used extensively to study injection, dispersion and combustion of liquid spray. Sun et al [77 investigated the atomization and vaporization of plain jet atomizer in a venturi tube, a candidate approach for application in ultra-low NOx combustors. Dodge and Scwalb [78] conducted detailed experimental investigation of a research swirl atomizer and showed that starting with measured information at an axial station sufficiently downstream from the nozzle exit, one can get fairly good agreement between predictions and measurements. A similar observation was made by Bulzan et al [79] with another air-assist airblast nozzle design. The effect of ambient pressure on the atomization and spray distribution has been investigated in References 80 and 81 for the research swirl atomizer and a practical fuel injector, respectively. The investigation on another practical liquid fuel injector with propane as reported by Hedman et al [82], and Schmidt and Hedman [83] has clearly shown the complexity of the process of flame stabilization, mixing and heat release. The fundamental studies reported by Nakabe, Mizutani and Hirao [84] on the burning characteristics of premixed sprays and gas-liquid coburning mixtures; Trichet, Lavergne and Biscos [85] on amplitude and temporal response of droplets in turbulent flow field; Chakravarty, Yang and Richards [86] on the droplet-vortex interactions in a turbulent round jet; and Li, Libby and Williams [87] on spray structure in two-phase counterflowing streams with and without a flame have provided valuable information to help improve the fuel-air mixing in practical devices. Finally, the flame stabilization and lean blowout processes, as measured by Sturgess et al [88], Sturgess and Shouse [89], and Switzer et al [90], clearly show the inadequacy of current mechanistic models.

As indicated in the above paragraph and by Bachalo [76], even though the phase Doppler interferometer (PDI) has been used extensively for investigating complex two-phase flows, detailed measurements on production hardware is less prevalent. A series of studies have been conducted on hardware developed by GE Aircraft Engines at UCI and AFWL as summarized in References [91-97]. The time averaged picture of the flow structure from the measurements or CFD calculations [47] is somewhat misleading in that the flow is actually highly dynamic in the time resolved picture as illustrated in Wang et al [93, 94]. The hardware scaling studies [95] revealed that, while the gas phase mean and fluctuating velocities scaled well, the droplet behavior did not since the atomization mechanism present in the simplex atomizers utilized did not produce drop sizes which scaled with basic nozzle descriptors (e.g., flow number). Examples of other studies include limited parametric studies of geometry, e.g., spray angle, flare expansion angle [92]. In these examples, it was found that spray angle has little effect on the droplet distribution or gas phase structure for the case studied whereas the flare exit had a significant effect on both gas and droplet behavior. An increase in large droplets in the center of the spray with the wide flare seemed to correlate with increased CO levels along the center of the combustor in that case.

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In addition, consideration of reacting flows produced with the swirl cup have been reported [97], as well as measurements for cases at proper inlet temperatures and pressures [96]. In both these cases, similar structure was found for both the reacting and non-reacting flows. A critical difference was that the reacting tests were conducted in a confined environment which has a significant effect on the gas phase structure, especially the swirling velocities.

The above studies, while a reasonable start in providing data for CCD challenges have some significant shortcomings. First, the complex interaction between the two counter-swirling air streams and the liquid filming process is not properly characterized due to the lack of optical access. As a result, key physics associated with the mixing process are not documented. Second, the information obtained is time averaged. Little is understood about the complex dynamic coupling between the phenomena occurring. Third, the parametric variations have been too limited to provide useful correlations or necessary insight into the effect of geometry or operating conditions. Finally, the studies have largely documented the flow structure without regard for performance issues. Whether or not the spray angle affects the aerodynamics may not be relevant in the absence of performance data (e.g., emissions, temperature patterns, etc.).

Recognizing the complexity associated with production fuel/air mixing devices, simpler test configurations that capture the essential processes in controlled manner need to be investigated. A good example, from industry's perspective is the study reported in References 98 and 99. An extensive velocity and temperature data base, including triple velocity correlations, has been compiled for double-concentric turbulent air jets and hydrogen jet diffusion flames, with or without swirl. Realizing this and for other practical reasons, GE Aircraft Engines GEAE) in partnership with four universities (the University of California campuses at Berkeley and Irvine, the University of Illinois at Urbana, and the Penn State) has started a joint experimental investigation for gas turbine combustor relevant simple flows, and the two generic classes of fuel/air mixing devices (namely axial and radial swirlers) sponsored under the AFOSR Focused Research Initiative (FRI). The main objectives of this four-year effort is:

A. Measure the spatial and temporal variations of flowfield variables including the three velocity components, mixture fraction, major species (CO_2 and unburned fuel, as minimum), intermediate combustion products (CO, as minimum), temperature, and NO_x.
B. Formulate and verify semi-analytical mechanistic models (SAM) for:
a) Recirculation zone characteristics (mass being recirculated, location, size, equivalence ratio and combustion efficiency, etc.).
b) Lean blowout and ignition.
c) Combustion efficiency, CO, UHC, NO_x and soot.
d) Radial profiles of major variables (axial velocity, swirl velocity, temperature, combustion efficiency, UHC, CO, NO_x.
e) Combustion dynamics input required for approximate methods and multi-dimensional non-linear unsteady flow calculations.

A parallel effort for the advanced model development is described in the following section.

Laurendeau and his associates [100-103] have applied LIF over the past five years to better understand NO formation in high-pressure gaseous flames. They have shown that point measurements of NO can be made up to at least 15 atm by employing narrow-band detection of NO fluorescence; and the feasibility in premixed flames down to 1 PPM of NO. This approach is being explored for application to lean premixed, partially premixed and diffusion flames using gaseous fuels up to 30 atm. GEAE is working with Laurendeau to extend the LIF approach for studying a lean-direct liquid fuel injection device.

The design challenges for maintaining exhaust smoke emissions well-below the threshold of visibility are becoming more demanding for the next-generation high-pressure low-emissions combustors. From the list of papers reviewed by Glassman [104] and Santoro, the combustion designer can appreciate the complexity of soot formation/oxidation processes, especially in regard to soot particle precursor chemistry, soot particle inception, surface growth and oxidation. Whereas these fundamental studies have improved the designers' insight, high-pressure investigations need to be taken starting from simpler flows and moving on ultimately to tackling the production devices in a manner somewhat similar to what UCI and AFWL have accomplished for studying the fuel-air mixing in GEAE's swirl cup[91-97].

Advanced CCD for Design

The failure of the second-order closure models of the 80's for predicting complex non-reacting flows [21, 39, 45, 47, 49, 51] has stalled the development of more accurate gas turbine combustor models for performance, emissions, liner wall temperatures and dynamics. Consequently, the combustion designer has to rely on hybrid modeling [52-61] and/or anchored approach [1] for making quantitative comparison with data. Recent turbulence modeling activities [106-116] appear to show promise enough for GEAE to investigate its adaptation in the design system.

The two most recent and extensive review by Faeth [117] and Sirignano [118] summarize recent progress made in the modeling of practical sprays, in addition to the topical review by Bachalo [76]. Of particular interest to us are the recent

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studies reported in References [79, 119-121]. These improvements in regard to droplet dynamics, heat-up and evaporation, and the enhancements due to the studies conducted by Raju [122] on the Monte Carlo PDF simulation of reacting sprays are in the process of inclusion into GEAE's combustion design system.

The challenging job of finding a reduced kinetic scheme that does not have too many species and yet does a reasonable job for simultaneously predicting NOx, CO and unburned hydrocarbons from idle to maximum power is no where close to completion [123-127]

The baseline flamelet approach provides as an extension of the fast chemistry to account for finite rate chemistry. However, the traditional flamelet approach still relies on the assumption that 1) the time scales for chemical kinetics are much shorter than both the time scales of convection and diffusion; 2) the flame thickness is smaller than the Kolmogorov length scale. Combustion generated pollutants, such as soot and NO_x, are formed near the reaction zone. Since the formation rates of pollutant species are relatively slow compared to combustion, the concentration of pollutant species depend strongly on the residence time. The flamelet assumptions made for combustion species is generally not applicable to pollutant species; consequently, the pollutant concentrations can not be obtained directly from the flame library [128].

One of the controversial issue in the flamelet approach is the choice of the parameter to describe the nonequilibrium nature of combustion [129]. It has been well known in laminar flames that NO_x formation is strongly influenced by the chemical kinetics. Two possibilities have been proposed in the past to quantify the nonequilibrium state of chemical kinetics. The traditional one is associated with the characteristic time of turbulent mixing and the alternative one is correlated to the fluid dynamics strain imposing on the flame. Although these two parameters are related to each other, their dependence on the turbulent flow field can be significantly different. Consequently, the predicted emissions will be highly sensitive to the choice of the nonequilibrium parameter. Some preliminary studies been conducted for turbulent hydrogen jet flames suggesting that the characteristic mixing time is more appropriate than the strain rate for predicting NO_x emissions [128].

Another issue in the flamelet model is related to the detailed molecular transport used in generating a flamelet library. The flamelet library consists of laminar flames that are computed using detailed molecular transport, and hence they contain full effects of preferential diffusion among heat and species. Recent experimental data on turbulent jet diffusion flames indicate that the effect of preferential diffusion depends on the jet exit velocity and on the axial location in turbulent jet flames [130-131] The flamelet approach tends to over-predict the preferential diffusion effects which can lead to inaccurate NO and CO predictions [132]. Issues like these need to be resolved before one can be reasonably sure for applying the laminar flamelet modeling with reduced kinetics to gas turbine combustors.

The Monte Carlo scalar and/or joint velocity/scalar PDF modeling approaches address many of the second-order closure modeling issues as summarized in References [133-135]. Consequently, the scalar and joint velocity/scalar PDF approaches have been applied [136] to a bluff body stabilized burner [137] and the CFM56 single annular gas turbine combustor, and compared with a specified PDF approach. All the three approaches gave similar results indicating that more work is needed to sort out the various issues, not the least of which is the computational time with the Monte Carlo simulation, before one can start using the latter in the design system. Considering the importance of computer and/or turn-around times in the design cycle, it is recommended to have a back-to-back comparison between the simulation techniques used by Pope [134] and Kollmann [138] adopted by Chen [128] and Raju [122].

The random vortex [139-143] and large-eddy simulations with linear eddy subgrid modeling [144-148] address many of the issues facing the laminar flamelet and Monte Carlo modeling approaches. Even though these advanced simulations currently may not be ready for solving the entire combustor flow field, its application for selected highly dynamic special-purpose problems is considered timely by GEAE.

The appropriate modeling of the dense part of a spray is crucial in determining the control parameters in the region most prone to soot and CO formation and oxidation in next-generation high-pressure ratio engines. Additionally, the evolution of the spray in the region most prone to NOx production depends upon the characteristics of the dense spray region. Many of these issues are being addressed by extending the dilute-spray Lagrangian formulations [118]. Results from the Eulerian-based models and numerical studies of clusters of drops [149-152] have predicted that it is precisely in the dense spray region that most of the control parameters can be found. These clusters of drops models account for drop interactions in terms of flow and drop dynamics, and limitations on heat and mass transfer. The current challenge is finding an approach to incorporate these cluster models as submodels into a spray code. The requirements are that the approach should both describe the crucial drop interaction in a turbulent flow and be economic to use for design purposes. The Eulerian-Eulerian approach must compromise on the time history of the drops but has the potential of a state-of-the-art description of interaction of groups of drops with turbulence. Monte-Carlo approaches can follow test drops and could, in principle, describe both drop interaction and interaction of drops with turbulence, although at great computational cost; their difficulty is in the "testing" of the number of drops to be followed in order to achieve a meaningful solution given the fact that this testing must be

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conducted, for example, every time changes in injectors, atomizers or fuels occur.

Summary

The gas turbine combustion design community has been making qualitative use of the recent fundamental understanding of aerothermal chemistry aspects of combustion, in addition to the computational combustion dynamics in its design process. Whereas this has been very useful in the past, mechanistically more accurate models are needed for designing the next-generation energy-efficient gas turbines with ultra-low emissions. The combustion research community and industry need to work closely and utilize advanced diagnostics and dynamically accurate turbulent combustion modeling approaches for the next-generation combustion design system.

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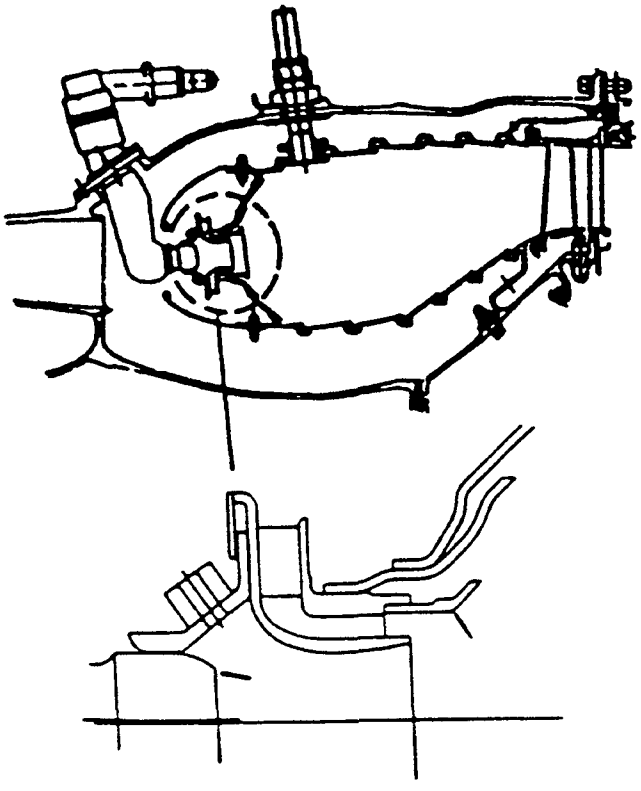


Fig. 1 A low-emissions single annular combustor for a nominal 30 OPR large commercial engine [1]

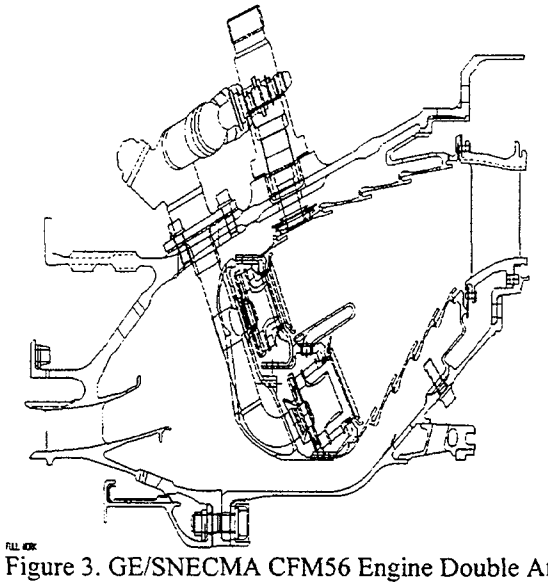


Figure 3. GE/SNECMA CFM56 Engine Double Annular Combustor (DAC) Configuration.[1]

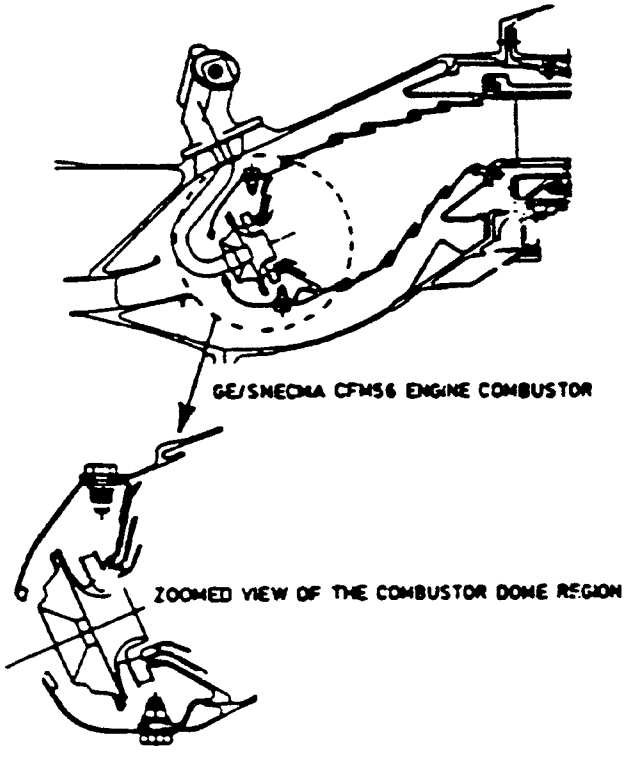
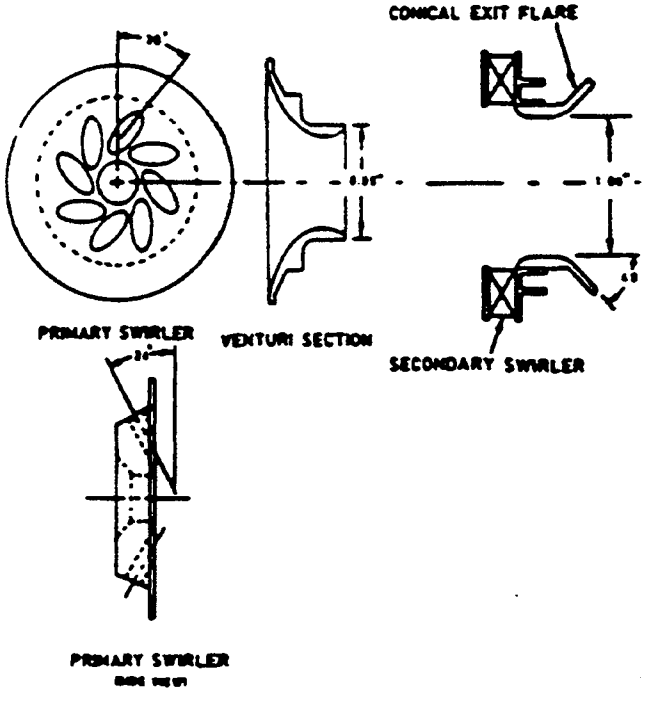


Figure 2. GE/SNECMA CFM56 Engine Single Annular Combustor (SAC) and details of the swirl cup [1].



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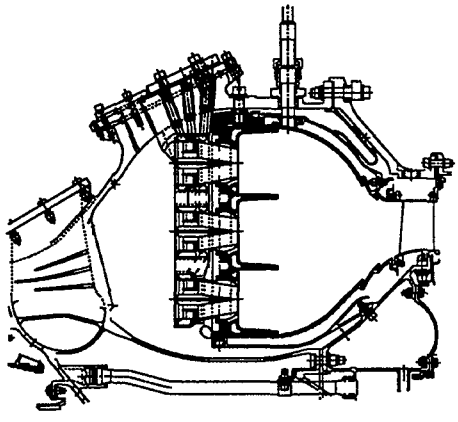

Fig. 4 Triple annular combustion system for ultra-low emissions aero-derivative engines[2]

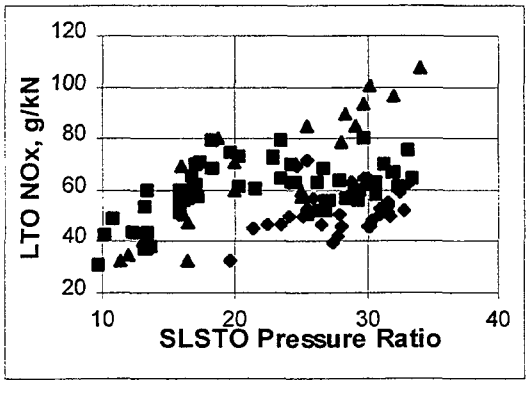

Fig. 5 Landing-Takeoff Characteristic NOx of large commercial turbofan engines.

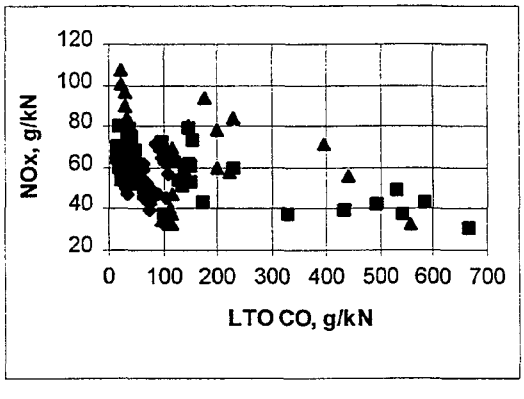

Fig. 6 Landing-Takeoff Characteristic NOx and CO of large commercial turbofan engines.

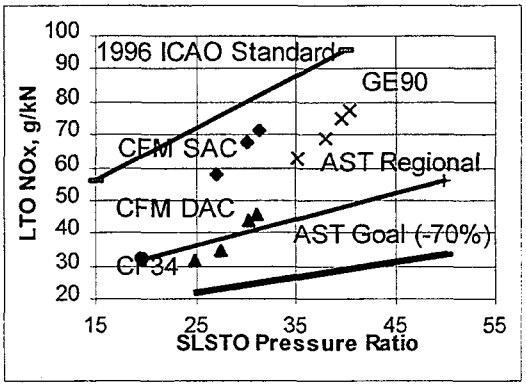

Fig. 7 Typical LTO NOx of single annular (SAC) and dual-annular (DAC) combustors.

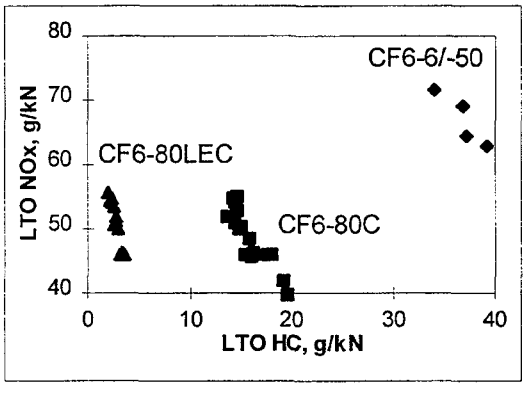

Fig. 8 Evolution of NOx/HC design tradeoffs for typical single annular combustors (SAC).

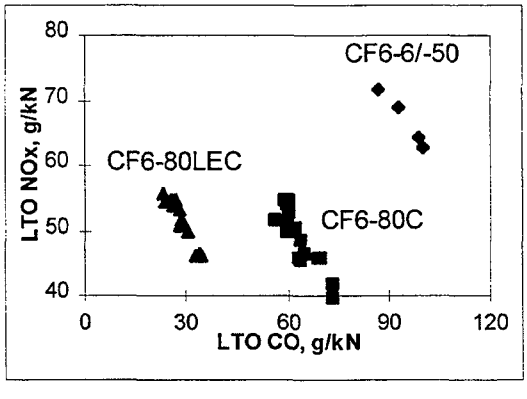

Fig. 9 Evolution of NOx/CO design tradeoffs for typical single annular combustors (SAC).

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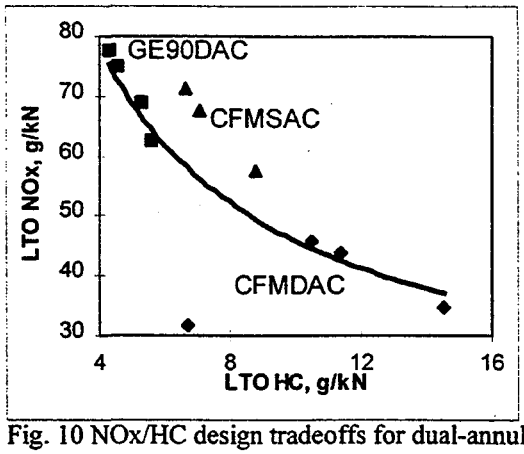

Fig. 10 NOx/HC design tradeoffs for dual-annular combustors (DAC) as compared with the CFM single annular combustor.

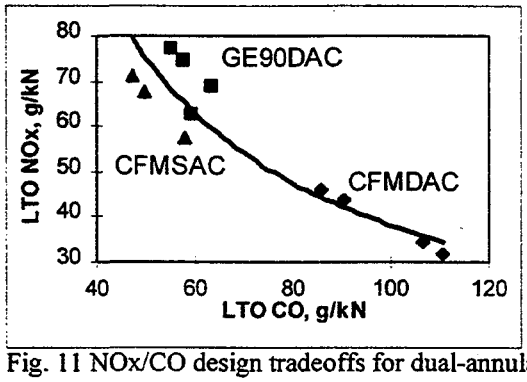

Fig. 11 NOx/CO design tradeoffs for dual-annular combustors (DAC) as compared with the CFM single annular combustor.

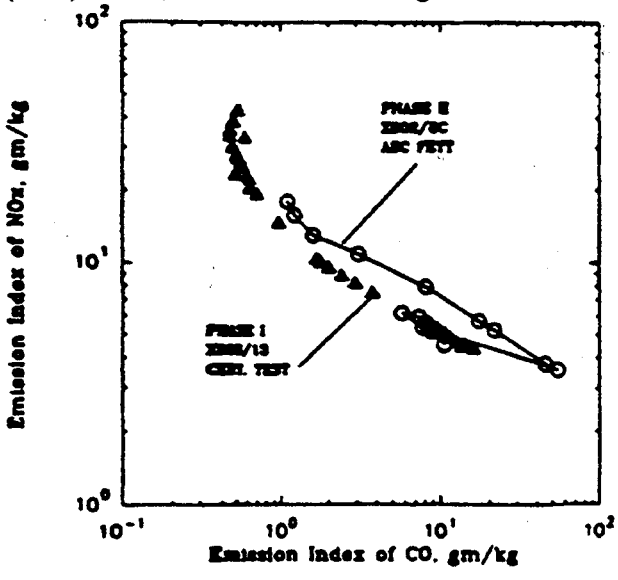

Fig. 12 NOx/CO design tradeoffs for axially-staged combustor (ASC) as compared with the single annular combustor Phase I from Segalman et al [5].

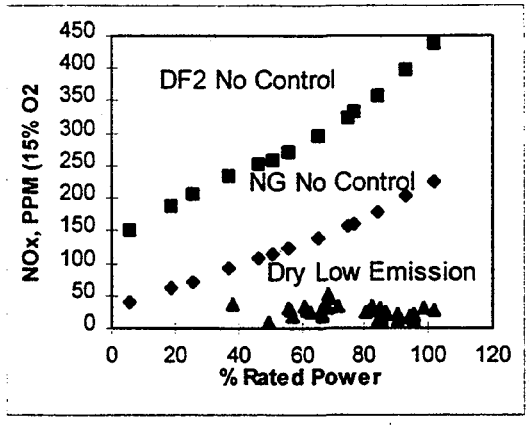

Fig. 13 Typical NOx emissions from a nominal 30 pressure ratio aero-derivative industrial engine with natural gas and DF-2.

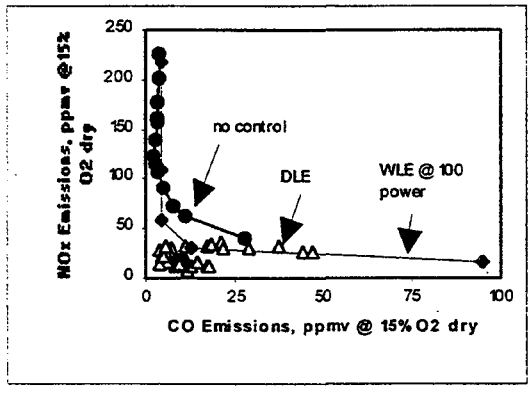

Fig. 14 Typical NOx/CO emissions characteristics from a nominal 30 pressure ratio aero-derivative industrial engine with natural gas.

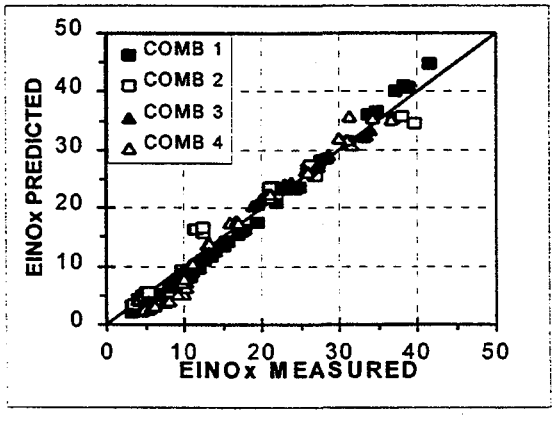

Fig. 15 Typical comparison between predicted and measured NOxEI from four modern propulsion engine combustors[17].

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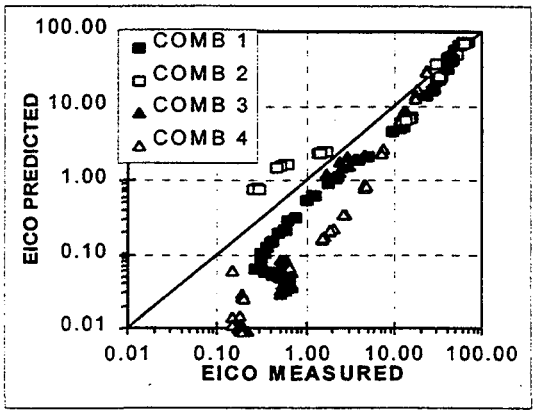

Fig. 16 Typical comparison between predicted and measured COEI from four modern propulsion engine combustors[17].

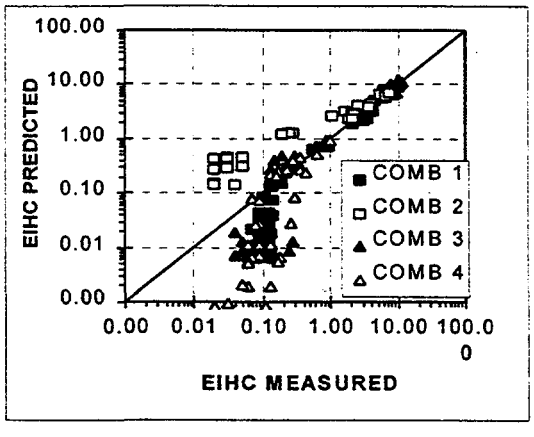

Fig. 17 Typical comparison between predicted and measured COEI from four modern propulsion engine combustors[17].

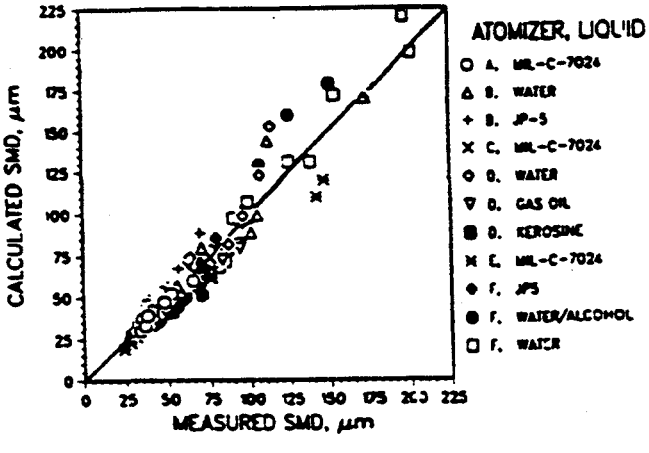

Fig. 18 Typical comparison between predicted and measured SMD from several airblast injectors from Rizk and Mongia [18].

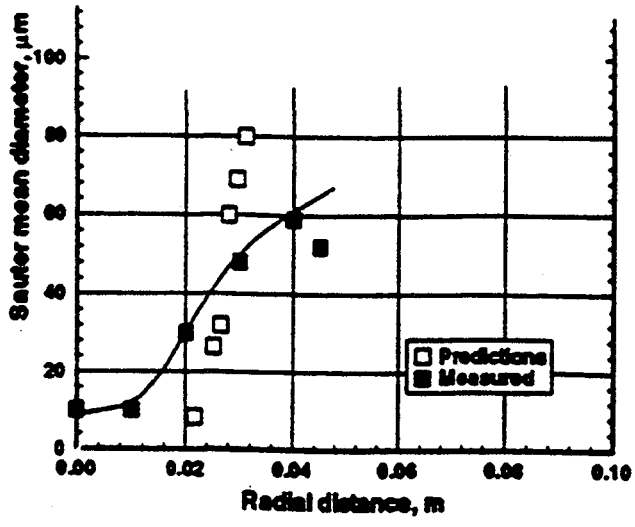

Fig. 19 Typical comparison between measured and predicted radial profiles of SMD 50 mm downstream of a reacting airblast nozzle [46].

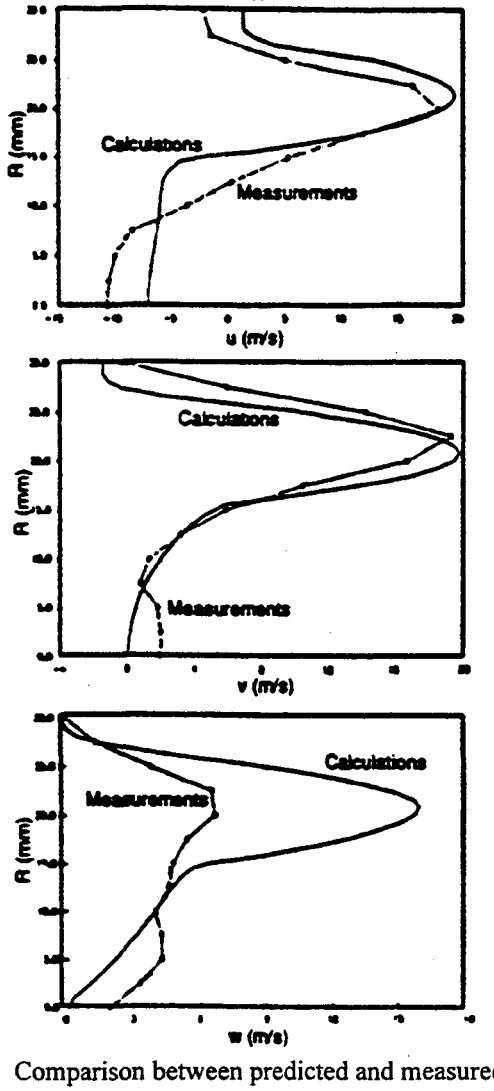

Fig. 20. Comparison between predicted and measured radial profiles of mean axial, radial and swirl velocity components 17 mm downstream of swirl cup [47].

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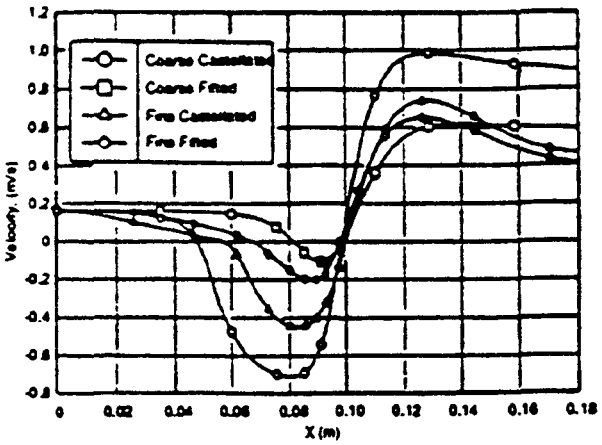

Axial Velocity Profiles Along the Centerline

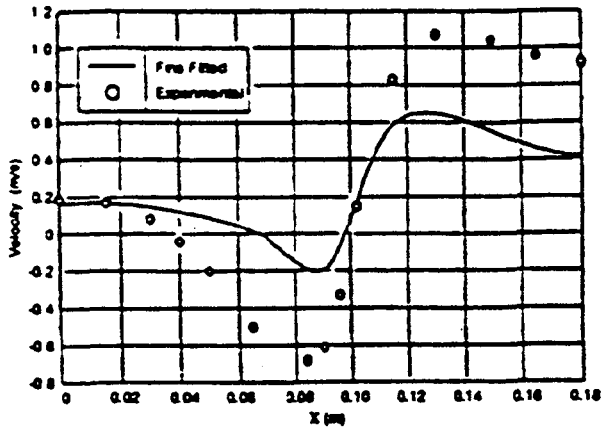

Fig. 21 Effect of dilution orifice simulation and mesh size on predicted axial velocity profiles along the jet centerline [48].

Table I: Accuracy goals for next-generation gas turbine combustion models - approximate and CCD-based.

Variable	Predictive accuracy band
NOx	± 5 %
CO and unburned hydrocarbons	± 10 %
Exhaust SAE Smoke Number	± 2
SLS idle lean blowout fuel/air ratio	± 0.001
Pattern factor (Peak Profile Factor)	± 0.03
Average radial profile factor	± 0.015
Maximum liner wall temperature	± 25° F
Combustion system pressure drop	± 0.25 (% Pt3)
Diffusion system pressure drop	± 0.125 (% Pt3)
Circumferential location of the liner hot spot	± 0.1 × Nozzle sector
Axial location of the liner hot spot	± 0.01 × Comb. length

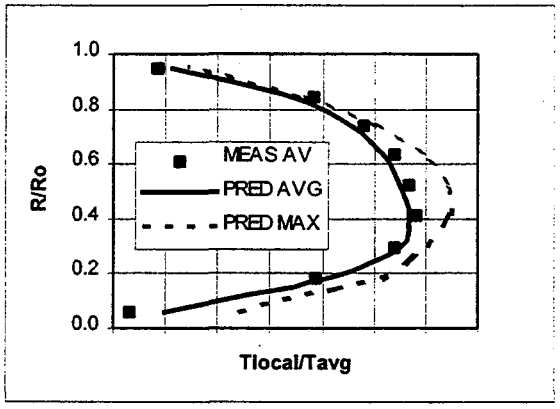

Fig. 22 Comparison between predicted and measured radial profile of exit temperature for the Anchored Combustor .

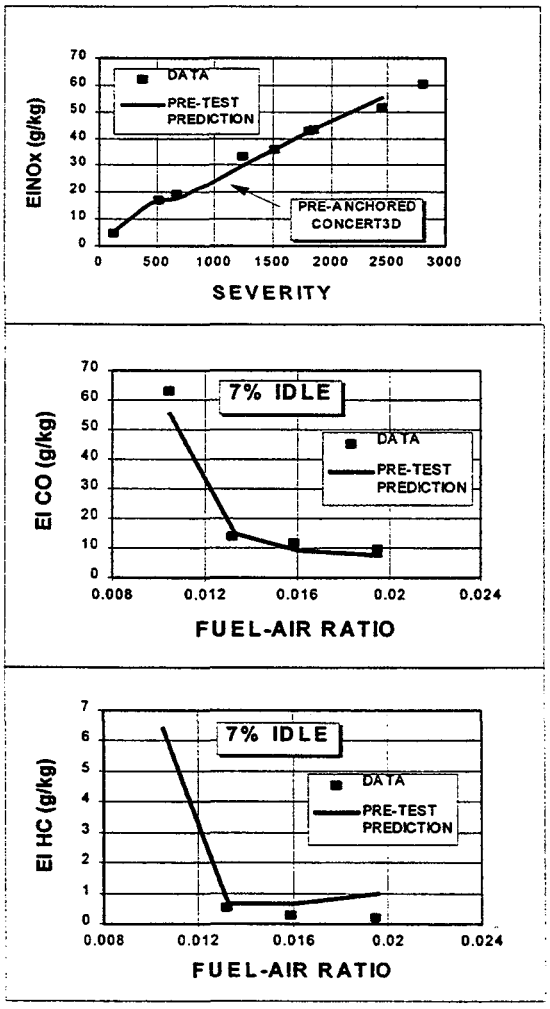

Figure 23. Comparison between pre-test predictions and measured emissions characteristics of the Designed Combustor .

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TABLE II. EMISSIONS/LBO/EXIT TEMPERATURE PREDICTION AGREEMENT.

	POWER RANGE*	EINOx	EICO	EIHC	LBO	RPF	PTF
Combustor 5	ALL	25%	69%	62%			
ANCHORED	HIGH	8%	1.5 EI	0.4 EI		6%	7%
CONCERT	LOW		4%	65%	10%		
COMB 1-4	ALL	35%	51%	59%			
ANCHORED	HIGH	13%	0.5 EI	0.2 EI		24%	9%
CONCERT	LOW		43%	34%	14%		
Combustor 6	ALL	24%	42%	39%			
ANCHORED	HIGH	12%	0.8 EI	0.2 EI		11%	4%
CONCERT	LOW		7%	10%	N/A		
DRA for Emissions from Danis, Pritchard and Mongia (1996)							
COMB 1-4	ALL	15%	56%	56%			
SINGLE	HIGH	8%	0.85 EI	0.15 EI			
REACTOR	LOW	23%	34%	27%			

* NOTE: HIGH POWER = NOx > 10 EI; LOW = NOx < 10 EI